Interface Lab

Session: IMA Low Res, Summer 2020
Code: IMALR-GT 1004 (2 credits)

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Course Description

This production course will survey alternative interfaces with an emphasis towards embodied interactions. Incorporating aspects of physical and tangible computing, students will be exposed to the internal machinations of systems, networks and sensors that underlie these interfaces. Areas covered include microcontrollers, connected devices, computer vision, virtual reality and augmented reality amongst others.

Course Objectives

At the completion of this course students will be able to demonstrate basic understanding of:

- Design concepts that incorporate software and hardware elements in interfaces
- Unity 2D and 3D scene design and asset management
- C# scripting to connect variables among different objects
- Physical computing elements such as sensors and actuators
- Arduino basic syntax to manipulate variables
- Communication concepts
- Electrical diagrams to connect components in a breadboard

Course Outline

Week 1

Prereqs:

- Download and join the [Discord] and say hello
- Download and Install Unity Hub and the latest release of Unity (2019.4 as of the date of writing)
- Download and install the latest release of Arduino IDE (Link)
- Get familiar with IOT nano 33 (Link)

Self-Paced Activities:

- Unity Tutorial Series [videos]: The Very Basics
- Arduino Tutorial Series: Electronics
- Introduction to Microcontrollers [video + readings]

Live Session:
- Game Engines as the Future of Interactive Media Interfaces
- Imagine: A Digital Interface to Everyday Things
- Hello Arduino & Paper Prototyping

Readings & Videos [optional]:
- [Video] Zach Lieberman: Interactive Art
- [Video] Massimo Banzi: How Arduino is open-sourcing imagination
- Programming Terms and Programming Environments
- Getting Started in Electronics by Forrest M. Mims
- Understanding the Breadboard
- Culkin, Jody, and Eric Hagan. Learn Electronics with Arduino : An Illustrated Beginner's Guide to Physical Computing. Chapter 2: Your Arduino, Maker Media, Inc, 2017. Link
- ITP Electronics Lab
- Eric Forman's Hello Circuits
- Prototyping Guidebook by Andy García

Project: Virtual Sculpture
Using the freeform example in the video as a guide, create a *virtual sculpture* - you can choose to have the sculpture dynamic (e.g., when pressing play things happen) or static. Make use of at least 2 different primitive 3D shapes, 2 materials and 2 lights.

Take a screenshot of your scene, or if you have a dynamic scene, record a video (using OBS or Quicktime) and upload it to the class Discord.

Project: Physical Interface (alpha version)
Using basic prototyping tools, including the Arduino board, build a physical device that is a representation of the imaginary interface we designed in class. The interface should have LEDs, potentiometers and push buttons. **Don't worry about making it actually function!** Instead, focus on the affordances that the interface provides. Record and upload a short video to the class Discord demonstrating your creation by the next live session.

Week 2

Self-Paced Activities:
- Unity Tutorial Series [videos]: Input and Scripting
- Arduino Tutorial Series: Sensors & Conditionals
- Arduino-Unity Tutorial Series: Communication

Live Session:

- Serial Communication in Unity: Ardity
- Unity + Arduino Mashups

Readings & Videos [optional]:

- Microcontrollers: The Basics
- Art+Science NOW. Chapter 6 “New Interfaces” by Steven Wilson
- Söke Dinkla, “The History of the Interface in Interactive Art” (1994)
- Crawford, Chris. The art of interactive design (Chapter 13) – NYU Library code QA76.76.I59 C73 2003
- Programming Terms and Programming Environments

Project: One Button Interface

Using the work of Week 1 as a source of inspiration, connect a button to an Arduino as the sole input you have to control one or more object(s) in your scene (this can be a shape, a light, a material). Your goal can be as simple as changing the position of a 3D shape, or as complex as changing the sky from day to night. What’s important is that you are able to receive input from the physical world (arduino) and *do something* with that in the virtual world (unity)

Think about creative ways to respond to different ways the button can be used (pressed, released, held, etc…). In addition, challenge yourself to find creative ways to create a button (for example

Week 3

Self-Paced Activities:

- Unity Tutorial Series [videos]: A simple 2D game
- Arduino Tutorial Series: Working with Many Sensors

Live Session:

- Let’s play: One Button Interfaces
- Sensor fusion from Arduino to Unity

Readings:

- Tom Igoe & Dan O’Sullivan, Physical Computing: Introduction
- A Brief Rant on the Future of Interaction Design by Bret Victor
- Responses: A Brief Rant on the Future of Interaction Design by Bret Victor
- Design Meets Disability (available at NYU SH library, code NK1110 .P85 2009)
- The Design of Everyday Things, Donald A. Norman (Section “The Seven Stages of Action”)
- Making Interactive Art: Set the Stage, Then Shut Up and Listen by Tom Igoe

- The Art of Interactivity by Margaret Morse

Project: Timely feedback
Build upon the simple game we created in the video series, only this time, incorporate *feedback* to let the player know something has happened. This should be non-screen-based feedback using one or more LEDs. For example, you might want to light up a red LED if the player hits a wall, or perhaps you want to blink the LED each time the player jumps. You can also incorporate other sensors if you'd like!

Week 4

Self-Paced Activities:
- Unity Tutorial Series [videos]: Audio
- Arduino Tutorial Series: Working with Servos
- Arduino-Unity Tutorial Series: Bidirectional Communication

Live Session:
- Let's play: Timely Feedback
- Introduction to Spatial Audio
- Introduction to Actuators

Readings & Videos [optional]:
- Arduino's Servos Library: Optional Commands - Makezine
- How to Make Almost Anything: The Digital Fabrication Revolution by Neil Gershenfeld
- Haptic Techniques for Media Control
- Experience Prototyping
- Barbie Liberation Organization
- Video: Norman Doors
- Hello Circuits: Art

Project: Mixed Reality or the Virtual-Physical Continuum
Using Unity's spatial audio system and your newly acquired ability to drive servo motors with Arduino, create a physically-spatially aware musical experience. You must use at least one sensor and one actuator in your project.

Week 5

Self-Paced Activities:
- Unity Tutorial Series [videos]: Interface Design
- Arduino Tutorial Series: Optimizing Execution Time

Live Session:
- Let's play: Spatial Instruments
- Imagine: Digital Physical Wonderland

Project: Digital Physical Wonderland

In your final project, we'll revisit the imaginary digital physical interfaces from our very first session, only this time, you'll be tasked with working together to prototype and demo one.